

Comparative Life Cycle Sustainability Assessment of Battery Energy Storage Systems and Pumped Hydro Energy Storage

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Executive Summary

This Life Cycle Sustainability Assessment (LCSA) compares lithium-ion Battery Energy Storage Systems (BESS) and Pumped Hydro Energy Storage (PHES) across environmental, economic, and social sustainability categories. Both systems were evaluated using a cradle-to-grave framework normalized to the delivery of 1 MWh of electricity to the grid over the system lifetime.

The Environmental LCA showed that BESS systems are dominated by upstream impacts from lithium, cobalt, nickel, and graphite extraction along with energy-intensive battery manufacturing (IEA, 2023; UNEP, 2011). These stages contribute heavily to climate change, resource depletion, water use, and human toxicity impacts. PHES systems instead concentrate impacts in the construction phase through concrete and steel production, excavation, reservoir creation, and ecosystem disruption (Yang & Jackson, 2011). However, PHES systems generally operate for much longer lifetimes with lower operational emissions.

The Life Cycle Costing analysis found that PHES systems generally have lower long-term costs for large-scale, long-duration storage because they avoid frequent replacement cycles and distribute infrastructure costs across decades of operation (IRENA, 2022). BESS systems are easier to deploy and less constrained by geography, but battery degradation and replacement substantially increase long-term costs (Zakeri & Syri, 2015).

The Social LCA identified different stakeholder burdens for each technology. BESS systems are tied to ongoing mining and supply-chain concerns involving worker safety, environmental justice, and material sourcing in regions with inconsistent labor protections (Amnesty International, 2016). PHES systems create more localized social impacts during construction through land transformation, reservoir development, and possible community displacement.

Neither technology was consistently preferable across all sustainability categories. PHES performed better for long-duration storage where suitable geography exists and long project lifetimes are achievable. BESS performed better in modular, flexible, and location-constrained applications. The results depended heavily on project location, operational requirements, electricity mix assumptions, and stakeholder priorities.

This assessment relied primarily on secondary literature sources and generalized assumptions. The findings should therefore be interpreted as a comparative high level study rather than a definitive ranking of either technology.

Goals & Scope

Goal Statement

The goal of this Life Cycle Sustainability Assessment (LCSA) is to compare the sustainability of lithium-ion battery energy storage systems and pumped hydro energy storage systems for large-scale grid electricity storage applications in the United States. The assessment evaluates environmental, economic, and social impacts across each system's life cycle in order to compare the tradeoffs associated with grid-scale energy storage deployment.

The study was conducted in the context of increasing renewable electricity deployment and the growing need for reliable storage technologies capable of balancing intermittent power generation from solar and wind resources. Utility-scale storage systems are used for load balancing, renewable integration, peak shaving, and grid reliability, but different storage technologies introduce different environmental burdens, infrastructure demands, operational constraints, and social impacts (IRENA, 2020; U.S. DOE, 2023).

This project does not attempt to identify a universally “best” storage technology. Instead, it compares where each system performs well, where each system creates larger burdens, and how those tradeoffs shift depending on geography, project scale, operational lifetime, and stakeholder priorities. The intended audience includes students, educators, and readers interested in sustainability assessment methods, energy systems, and infrastructure planning.

The study also demonstrates how LCSA methods can be applied to engineering systems where environmental, economic, and social outcomes do not align cleanly. Because of this, the analysis places strong emphasis on assumptions, system boundaries, data quality, and uncertainty.

Scope Statement

Systems to be Studied

The systems studied in this report are:

1. Utility-scale lithium-ion battery energy storage systems (BESS)
2. Pumped hydro energy storage (PHES) systems

The lithium-ion battery system evaluated in this report mainly reflects modern utility-scale lithium-ion battery storage technologies commonly paired with renewable energy generation and grid stabilization applications. The pumped hydro system reflects conventional closed-loop or reservoir based pumped hydroelectric energy storage installations used for large-scale grid storage.

The study compares the systems across environmental life cycle assessment (E-LCA), life cycle costing (LCC), and social life cycle assessment (S-LCA) categories.

Functional Unit

The functional unit used in this analysis is:

The delivery of 1 MWh of stored electricity to the electrical grid over the operational lifetime of the system.

Using this unit allows both technologies to be compared on the basis of equivalent delivered energy storage performance despite differences in infrastructure scale, operating lifetime, efficiency, and deployment characteristics.

System Boundary

The system boundary for both technologies is cradle-to-grave and includes the major life cycle stages for each storage system.

For lithium-ion battery storage systems, the system boundary includes:

- Raw material extraction and processing (including lithium, cobalt, nickel, graphite, and other battery materials)
- Battery manufacturing and assembly
- Transportation and installation
- Operational use and maintenance
- Battery degradation and replacement considerations
- End-of-life processing, recycling, and disposal

For pumped hydro storage systems, the system boundary includes:

- Raw material extraction and construction material production
- Civil infrastructure construction (reservoirs, tunnels, turbines, pumps, and supporting structures)
- Transportation and installation
- Operational use and maintenance
- Long-term infrastructure operation
- Decommissioning and end-of-life considerations where applicable

The analysis focuses mostly on utility-scale deployment in the United States context. However, some supply chain impacts (specifically for lithium-ion battery materials) extend internationally due to the global nature of mineral extraction and battery manufacturing.

Several limitations impact this study. Site-specific ecological impacts, regional policy differences, evolving battery chemistries, and future technological improvements can greatly influence results but can't be fully captured within the scope of this assessment. Additionally, some social and environmental impacts rely on secondary data sources and generalized assumptions because of limited publicly available information.

Despite these constraints, the selected scope provides a solid framework for comparing the relative sustainability implications of the two energy storage technologies across environmental, economic, and social facets.

Life Cycle Analyses

Environmental LCA

Phase 2, Life Cycle Inventory

Diagrams

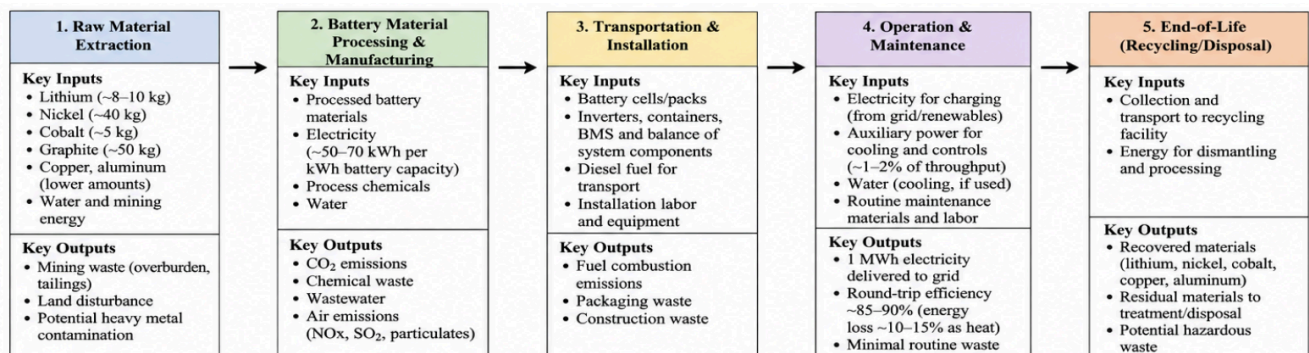


Figure 1: Life Cycle Inventory Diagram for Lithium-Ion Battery Energy Storage System
Functional Unit: 1 MWh of electricity delivered to the grid

Note: Quantities of materials are approximate and vary by battery chemistry and system design; values reflect typical ranges for grid-scale lithium ion systems.

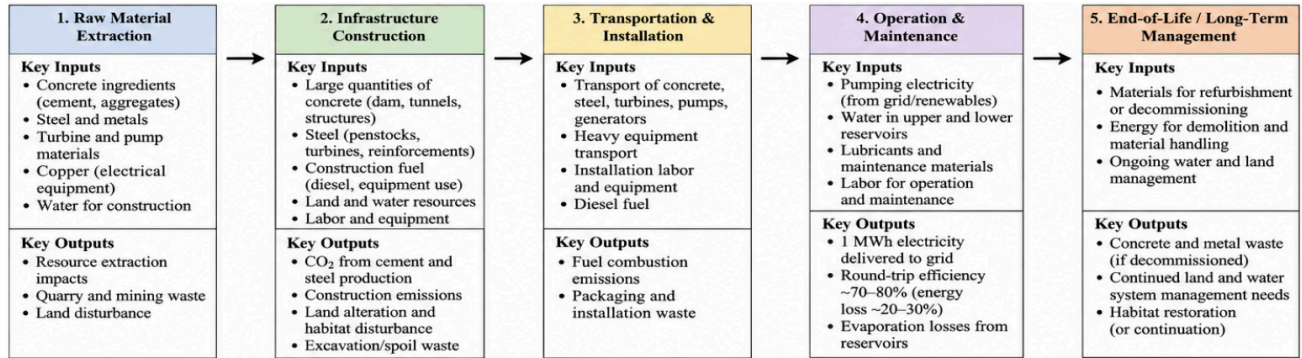


Figure 2: Life Cycle Inventory Diagram for Pumped Hydro Energy Storage System
Functional Unit: 1 MWh of electricity delivered to the grid

Note: Quantities depend on site specific design (head, storage volume, distance) and system efficiency.

Narrative Description of Inventory

Lithium-Ion Battery Energy Storage System (BESS)

The lithium-ion battery inventory was developed using a cradle-to-grave framework normalized to 1 MWh of delivered electricity over the operational lifetime of the system. The inventory includes raw material extraction, battery manufacturing, transportation, installation, operation and maintenance, and end-of-life management.

The raw material extraction stage includes mining and refining of lithium, cobalt, nickel, graphite, copper, aluminum, and other battery materials. These processes require large energy inputs, water consumption, and chemical processing operations. Mining activities also generate tailings, land disturbance, and greenhouse gas emissions.

Battery manufacturing represents one of the most energy-intensive life cycle stages because of electrode processing, cell assembly, thermal management production, and pack integration. Manufacturing electricity demand is particularly influential because many battery production facilities still operate in regions with fossil-fuel-intensive electricity mixes.

Transportation and installation impacts include international shipping of raw materials and finished battery packs prior to deployment in utility-scale storage facilities. Transportation contributes to fossil fuel consumption and emissions through marine freight and truck transportation, although these impacts remain smaller than manufacturing-stage burdens.

During operation, BESS systems experience round-trip efficiency losses, thermal management energy demand, and gradual degradation over time. Depending on operational

cycling conditions and project lifetime assumptions, battery replacement may be required during operation.

End-of-life management includes recycling, disposal, and material recovery. Recycling can reduce future virgin mineral demand, although recycling infrastructure for utility-scale lithium-ion systems is still developing and varies substantially by region and battery chemistry.

Several assumptions were required because utility-scale inventory data remain inconsistent across manufacturers and battery chemistries. Many inventory values were generalized from published grid-scale battery studies.

Pumped Hydro Energy Storage (PHES)

The pumped hydro energy storage inventory was also evaluated on a cradle-to-grave basis normalized to 1 MWh of delivered electricity over the system lifetime. The inventory includes infrastructure material production, construction, operation and maintenance, and end-of-life considerations.

The largest environmental burdens for PHES systems happen during construction. Major material inputs include concrete, steel, excavation materials, turbines, pumps, pipelines, and reservoir infrastructure. Construction activities require heavy equipment use, land modification, and carbon-intensive cement and steel production. Reservoir development can also alter habitats, water flow patterns, and surrounding ecosystems depending on project location.

Transportation and installation impacts include movement of heavy construction equipment and infrastructure materials to the project site. Similar to the BESS system, transportation impacts were treated as secondary relative to infrastructure construction burdens.

Operational impacts for PHES systems are comparatively low because electricity storage occurs through gravitational water storage rather than electrochemical reactions. However, operational losses still occur because of pumping inefficiencies and evaporation losses. The long operational lifetime of PHES systems strongly affects the inventory because infrastructure impacts are distributed across decades of operation and very large cumulative electricity throughput.

End-of-life impacts are difficult to quantify because many pumped hydro facilities continue operating beyond originally intended lifetimes. Decommissioning may involve partial infrastructure removal, continued reservoir use, or rehabilitation depending on geographic and regulatory conditions. Because PHES systems vary greatly by geography, reservoir design, and local environmental conditions, the inventory reflects generalized utility-scale systems rather than a specific installation.

Phase 3, Life cycle impact assessment

The life cycle inventory results showed that several midpoint impact categories from the UNEP LCSA framework are strongly affected by both energy storage technologies.

For lithium-ion battery systems, the most important midpoint categories include climate change, mineral resource depletion, human toxicity, ecotoxicity, freshwater use, and fossil fuel depletion. The extraction and refinement of lithium, cobalt, nickel, and graphite require energy-intensive mining and chemical processing operations that contribute heavily to greenhouse gas emissions and environmental contamination. In addition, mining activities may generate water pollution, soil degradation, and ecosystem disturbance. Battery manufacturing also contributes significantly to climate change impacts due to the large electricity demand associated with cell production and assembly.

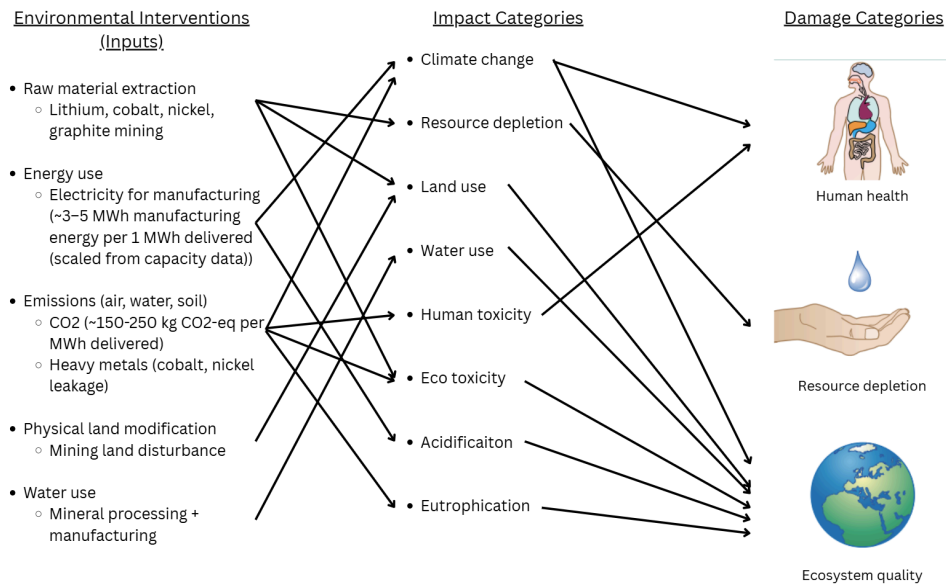


Figure 3. Relationship between environmental interventions, midpoint impact categories, and endpoint damage categories for the lithium-ion BESS system.

For pumped hydro systems, the dominant midpoint categories include land use, ecosystem disruption, water resource impacts, climate change from construction materials, and resource depletion associated with concrete and steel production. The construction phase contributes substantially to greenhouse gas emissions because cement manufacturing is highly carbon intensive. Reservoir construction and land modification may also alter habitats, water flow patterns, and surrounding ecosystems. However, because pumped hydro systems typically operate for very long lifetimes with relatively low operational emissions, many impacts become distributed across large cumulative electricity outputs.

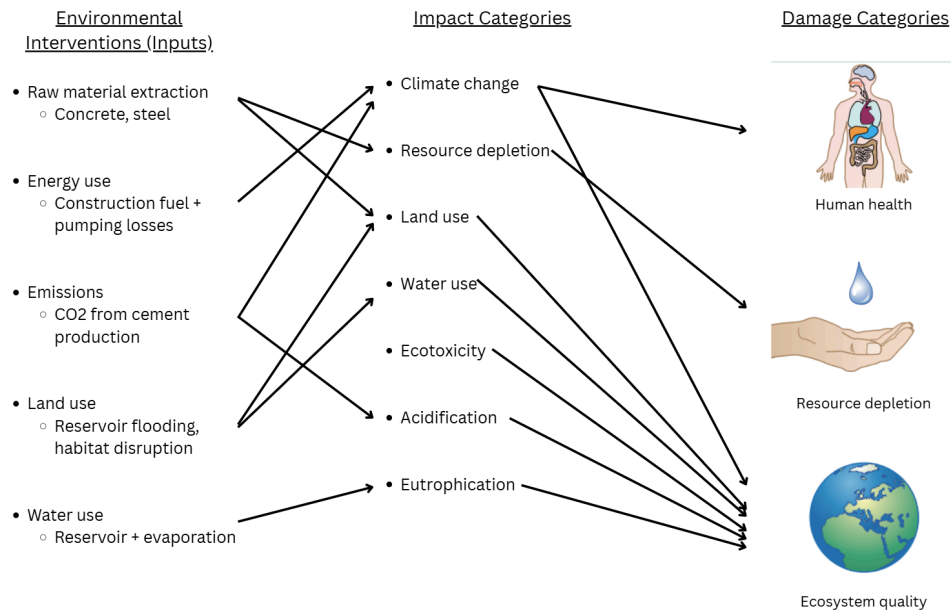


Figure 4. Relationship between environmental interventions, midpoint impact categories, and endpoint damage categories for the pumped hydro storage system.

Both systems contribute to climate change impacts through emissions in material production and infrastructure development, although the distribution of impacts differs substantially across life cycle stages. Lithium-ion systems are dominated by material extraction and manufacturing burdens, while pumped hydro impacts are concentrated primarily in construction and land modification stages.

At the endpoint level, these midpoint impacts affect human health, ecosystem quality, and resource availability. Human health impacts arise from mining-related contamination, air pollution exposure, and industrial emissions. Ecosystem impacts result from habitat disruption, land transformation, water contamination, and large-scale resource extraction.

Despite these concerns, both technologies still provide important environmental benefits relative to fossil fuel-based grid stabilization approaches because they enable increased integration of renewable electricity generation. As a result, environmental tradeoffs must be evaluated within the broader context of long-term energy system decarbonization.

Phase 4, Interpretation

Conclusions

The Environmental LCA does not support a clear conclusion that either system is environmentally preferable across all categories. BESS systems generally produce larger impacts related to mineral extraction, manufacturing energy demand, human toxicity, and resource depletion because lithium, cobalt, nickel, and graphite extraction require energy-intensive mining and processing operations. PHES systems instead produce larger impacts tied to land transformation, ecosystem disruption, construction emissions, and reservoir development.

The most influential life cycle stage for BESS systems is battery production. Mining and refining operations dominate climate change, water use, ecotoxicity, and resource depletion impacts (IEA, 2023). For PHES systems, the construction phase dominates environmental performance because concrete production, excavation, tunneling, and land modification generate large upfront environmental burdens (Yang & Jackson, 2011).

PHES systems benefit from long operational lifetimes and high cumulative electricity throughput, which lowers impacts per functional unit over time. BESS systems benefit from deployment flexibility and reduced geographic constraints, but their long-term environmental performance depends heavily on future battery chemistry development, recycling rates, and electricity grid decarbonization.

Completeness of the Study

Although the study captures major life cycle stages for both systems, there are still several areas of data uncertainty. Publicly available utility-scale inventory data for lithium-ion battery systems are limited and often vary substantially depending on battery chemistry, manufacturer, and operating assumptions. Similarly, pumped hydro environmental impacts are highly site dependent and difficult to generalize across all installations.

Additionally, future improvements in battery recycling, cleaner electricity generation, and evolving energy storage technologies could significantly alter long-term environmental outcomes. Operational performance assumptions such as round-trip efficiency, degradation rates, and lifetime cycling behavior also introduce uncertainty into the normalized inventory results.

Limitations

This study contains several important limitations. First, the assessment relies heavily on secondary literature sources and generalized inventory assumptions rather than primary industrial

data. Second, environmental impacts were evaluated qualitatively or semi-quantitatively rather than through a complete process-based numerical LCA software model.

Another limitation is that geographic variability is not fully captured. Mining practices, electricity generation mixes, manufacturing methods, and environmental regulations differ significantly across countries and regions. Similarly, pumped hydro systems vary substantially depending on reservoir design, local ecosystems, and hydrological conditions.

Finally, because the analysis evaluates sustainability at a broad systems level, many localized environmental effects and future technological developments cannot be fully represented within the scope of this study.

Response to Review, ELCA

Comments	Responses
1) Peer comment: One comment noted that the environmental LCA had a strong goal and scope statement, a clear functional unit of 1 MWh, and easy-to-follow figures. However, they suggested improving the diagrams by more clearly connecting quantified inputs and outputs to midpoint impact categories and damage categories to provide a more comprehensive overview.	This was a useful and relevant comment because it highlighted that my original figures functioned more as impact pathway diagrams than as true life cycle inventory diagrams. In response, I revised the Environmental LCA structure so that the Phase 2 inventory section now includes updated life cycle inventory diagrams for both BESS and PHES systems organized explicitly by life cycle stage: raw material extraction, manufacturing/construction, transportation and installation, operation and maintenance, and end-of-life management. Each stage now includes quantified inputs and outputs normalized to the functional unit of 1 MWh electricity delivered. I also moved the original environmental intervention → midpoint impact → damage category diagrams into the Phase 3 impact assessment section, where they better support the discussion of midpoint and endpoint impacts. This restructuring improved alignment with UNEP (2011) guidance and clarified the difference between inventory analysis and impact assessment.
2) Peer comment: The reviewer commented	This comment was very relevant to the final

that the functional unit and quantitative system inputs/outputs were clearly presented and supported by sources. They also recommended adding diagrams more similar to Figures 2 and 3 from Valdivia et al. (2011), further justifying why transportation impacts were minor compared to manufacturing and construction, and expanding discussion of tradeoffs between the systems.	report requirements. In response, I redesigned the inventory diagrams to more closely resemble the process based structure used in Valdivia et al. (2011), with distinct life cycle stages and quantified flows associated with each stage. I also expanded the narrative discussion to better justify why transportation impacts were treated as secondary relative to battery manufacturing and PHES infrastructure construction. I added more explanation was added describing how manufacturing energy intensity, concrete production, mining activities, and long-term infrastructure requirements dominate the environmental burdens relative to transportation emissions. Finally, I strengthened the interpretation section by more explicitly discussing tradeoffs between the systems. The revised conclusion now explains that BESS systems generally display higher impacts from critical mineral extraction and manufacturing, while PHES systems exhibit larger land transformation and ecosystem impacts due to reservoir construction and long-term water management.
3) TA comment: The TA noted that the written narrative was detailed and assumptions were clearly stated, but the diagrams were organized more like Phase 3 impact diagrams rather than Phase 2 inventory diagrams. The comment recommended restructuring the diagrams to show quantified inputs and outputs by life cycle stage so that the figures and narrative more directly complement each other.	This was an important structural comment that directly influenced the organization of the Environmental LCA section. In response, I fully separated the inventory analysis and impact assessment components of the report. The revised Phase 2 section now focuses exclusively on life cycle inventory information, including quantified inputs and outputs organized by stage for both BESS and PHES systems. The original midpoint and damage category diagrams were relocated into Phase 3 to support the impact assessment discussion instead of the inventory analysis. This change improved consistency with the assignment instructions and clarified the progression from inventory flows to midpoint impacts and ultimately to endpoint damage

	categories. The updated diagrams also more clearly connect to the written inventory narrative and improve overall readability and organization.
4) Self comment: While revising the Environmental LCA section, I realized that my original diagrams emphasized conceptual environmental impacts more strongly than the actual life cycle process flows required in the inventory phase.	After reviewing the rubric and feedback comments, I recognized that the inventory analysis needed a clearer distinction between physical system flows and environmental impact interpretation. To address this, I redesigned the Phase 2 diagrams to focus specifically on process stages, quantified material and energy flows, operational losses, and end-of-life outputs. I also ensured that both systems consistently used the same functional unit and comparable life cycle stages to improve transparency between the two technologies. If additional time and data were available, I would further improve the inventory by incorporating more site-specific transportation distances, regional electricity mix assumptions, and sensitivity analysis for battery chemistry variations and PHES reservoir design characteristics.

Life Cycle Costing (LCC)

Phase 2 and 3, Inventory and aggregate costs by cost categories

The functional unit for this Life Cycle Costing (LCC) analysis is 1 MWh of electricity delivered to the grid from either a lithium-ion Battery Energy Storage System or a Pumped Hydro Energy Storage system over the operational lifetime of the storage technology. Using a common functional unit allows for direct comparison between the two systems despite their differences in infrastructure scale, lifetime, efficiency, and operational characteristics.

Life Cycle Stage	Cost Category	Estimated Cost per Functional Unit (USD/MWh delivered)	Description
Raw Material Extraction &	Lithium, nickel, cobalt, graphite	\$18–30	Includes extraction and processing of

Processing	mining and refining		critical battery minerals and associated energy/material costs
Manufacturing & Assembly	Cell production, battery pack assembly, electronics	\$35–55	Energy intensive battery manufacturing and balance-of-system components
Transportation & Installation	Shipping, installation labor, infrastructure setup	\$5–10	Transportation of battery packs and construction/installation activities
Operation & Maintenance	Cooling systems, controls, maintenance labor	\$8–15	Operational electricity losses, auxiliary systems, inspections, maintenance
Replacement Costs	Mid-life battery replacement/degradation	\$20–40	Capacity degradation and partial/full battery replacement over project life
End-of-Life Management	Recycling, disposal, hazardous waste handling	\$3–8	Collection, recycling, disposal, and material recovery costs
Total Estimated Life Cycle Cost		\$89–158/MWh delivered	

Table 1. Aggregate Life Cycle Costs for Lithium-Ion Battery Energy Storage System

Sources: International Energy Agency (2023); Lazard (2024); Schmidt et al. (2019).

Life Cycle Stage	Cost Category	Estimated Cost per Functional Unit (USD/MWh delivered)	Description
Raw Material Extraction	Concrete, steel, aggregate extraction	\$10–18	Material extraction for dams, tunnels,

			turbines, and reservoirs
Infrastructure Construction	Civil works, excavation, dam construction	\$40–70	Major infrastructure and construction-related expenditures
Transportation & Installation	Heavy equipment transport and installation	\$4–9	Transport of turbines, pumps, steel, and construction materials
Operation & Maintenance	Pumping energy, inspections, maintenance	\$6–14	Operational energy losses and long-term maintenance requirements
Long-Term Infrastructure Management	Reservoir management and refurbishment	\$5–12	Long-term infrastructure upkeep and rehabilitation
End-of-Life / Decommissioning	Dam rehabilitation or decommissioning	\$2–6	Site restoration and infrastructure disposal costs
Total Estimated Life Cycle Cost		\$67–129/MWh delivered	

Table 2. Aggregate Life Cycle Costs for Pumped Hydro Energy Storage

Sources: International Renewable Energy Agency (IRENA, 2022); Lazard (2024); Yang & Jackson (2011).

Narrative Discussion

The Life Cycle Costing analysis compares BESS and PHES systems using the functional unit of 1 MWh of electricity delivered to the grid over the operational lifetime of the system. Costs were organized by life cycle stage in order to identify where each technology concentrates its economic burdens.

For BESS systems, the largest contributors to total life cycle cost are manufacturing and replacement costs. Battery manufacturing requires large quantities of refined lithium, nickel, cobalt, graphite, and copper, as well as highly energy-intensive cell production processes (IEA, 2023). In addition, lithium-ion batteries experience gradual degradation over time, often requiring partial or full replacement during the operational lifetime of the system, particularly in

high-cycle grid applications (Schmidt et al., 2019). These replacement costs significantly increase long-term life cycle expenditures relative to systems with longer-lived infrastructure. End-of-life costs are comparatively smaller but still important because battery recycling and hazardous material management require specialized processing and transportation infrastructure.

For PHES systems, costs are concentrated much more heavily in the initial infrastructure construction phase. Large-scale dam construction, excavation, tunneling, reservoir creation, and turbine installation require substantial capital investment and long construction timelines (IRENA, 2022). However, once operational, PHES systems typically have lower replacement costs and significantly longer operational lifetimes than lithium-ion batteries, often exceeding 50 years. Operation and maintenance costs remain relatively moderate and are primarily associated with pumping losses, infrastructure inspections, reservoir management, and mechanical maintenance.

The comparison shows a clear economic tradeoff. BESS systems reduce siting constraints and can be deployed quickly in a wider range of locations, which makes them attractive for shorter-duration storage or urban deployment. PHES systems require suitable geography and much larger upfront infrastructure investment, but they generally become more cost-effective for long-duration storage over long operational lifetimes. Replacement-related costs account for roughly 20-40 USD/MWh for BESS systems, while PHES systems instead concentrate most costs in upfront construction expenditures.

The results also suggest that the system suitability depends heavily on application and location. BESS systems may be more economically favorable in urbanized regions, smaller-scale deployments, or areas lacking appropriate topography for reservoirs. PHES systems may become more cost-effective in regions where large-scale, long-duration storage is needed and where geographic conditions support reservoir construction. These findings are relevant to grid planners and policymakers evaluating long-term renewable integration strategies and energy infrastructure investment decisions.

Phase 4, Interpretation

Conclusions

The Life Cycle Costing analysis suggests that PHES systems generally achieve lower long-term life cycle cost per functional unit than BESS systems when evaluated across long operational lifetimes. Although PHES requires much larger upfront infrastructure investment, its long service life and relatively low degradation reduce replacement-related costs over time.

BESS systems reduce deployment barriers and provide greater siting flexibility, but manufacturing and replacement costs are still major economic burdens.

Cost outcomes also depend heavily on project context. Geographic suitability, operational duration, electricity market conditions, cycling frequency, and future battery material prices can all substantially change comparative system economics.

Completeness of Study

This study uses secondary data sources and generalized cost ranges rather than project-specific financial data. Actual costs can vary substantially depending on project scale, battery chemistry, financing structure, labor costs, regional electricity markets, permitting requirements, and site-specific construction conditions. In addition, many publicly available cost estimates aggregate categories differently, making direct comparisons between sources difficult.

The analysis also simplifies long-term operational uncertainty by assuming representative lifetime averages for replacement cycles, efficiency losses, and maintenance requirements. More detailed future studies could incorporate discounted cash flow analysis, sensitivity analysis, regional electricity pricing, and dynamic operational modeling to improve economic precision.

Assumptions & Limitations

The analysis assumes generalized grid-scale lithium-ion battery systems and generalized PHES infrastructure configurations rather than site-specific projects. Transportation distances, financing structures, labor variability, and regional permitting costs were not modeled explicitly. Future reductions in battery manufacturing costs or improvements in recycling infrastructure, may significantly alter comparative results over time.

Additionally, the study focuses primarily on direct life cycle costs and does not quantify broader societal or externalized economic impacts such as environmental remediation costs, land opportunity costs, or social equity considerations. As a result, the findings should be interpreted as a comparative high-level LCC analysis rather than a precise project feasibility study.

Response to Review, LCC

Comments	Responses
1) Peer comment: The reviewer noted that the cost breakdown and separation of replacement and disposal costs were especially strong. They suggested expanding the conclusions section to more directly discuss what the findings mean for planners and policymakers, including situations where one system may be preferred over the other	This was a very helpful comment because it identified an area where the analysis could connect more directly back to the intended audience and broader sustainability implications. In response, I expanded the interpretation section to discuss how system suitability depends on application, geography, deployment scale, and storage duration

depending on location or use case.	requirements. I added discussion explaining that BESS systems may be more favorable for shorter-duration or urban deployments where flexible siting is important, while PHES systems may become more cost effective for long term storage in regions with suitable topography. I also strengthened the policy relevance of the conclusions by explicitly connecting the results to long-term renewable integration planning and infrastructure investment decisions.
2) TA comment: The TA commented that the aggregate cost table was clean and easy to follow and tied effectively to the functional unit. However, the narrative could improve by more directly comparing the magnitude of costs across life cycle stages rather than discussing each category separately.	This comment was directly relevant to improving the clarity and analytical depth of the narrative discussion. In response, I revised the discussion section to more explicitly compare which life cycle stages dominate total costs for each system. The updated narrative now emphasizes that BESS costs are driven primarily by manufacturing and replacement, while PHES costs are concentrated in infrastructure construction. I also added clearer comparative language discussing how PHES systems generally shift costs toward initial capital investment while BESS systems distribute costs more heavily across operational replacement cycles. This revision improved the overall synthesis of the results rather than presenting each cost category independently.
3) Self comment: While revising the Life Cycle Costing section, I realized that the original draft focused heavily on reporting cost categories individually but did not fully explain how operational lifetime and system context influence comparative cost performance between the technologies.	To address this, I expanded the interpretation section to include discussion of system lifetime, cycling behavior, infrastructure durability, and geographic suitability. I also improved the comparison between short-duration and long-duration storage applications to better explain why the economic advantages of each system may vary depending on use case. If additional time and project-specific data were available, I would further improve the analysis by incorporating discounted cash flow modeling, regional electricity pricing, sensitivity

	analysis for battery replacement frequency, and uncertainty analysis for future battery material costs.
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Social Life Cycle Assessment (S-LCA)

Phase 2, Inventory

The Social Life Cycle Assessment (S-LCA) compares the social implications associated with lithium-ion Battery Energy Storage Systems (BESS) and Pumped Hydro Energy Storage (PHES) systems across their life cycles. Following the UNEP/SETAC (2013) methodological sheets for S-LCA, two impact subcategories were selected and mapped to their corresponding stakeholder categories in order to evaluate the major social tradeoffs associated with each energy storage technology.

The selected impact subcategories were chosen because they represent some of the most frequently discussed social concerns associated with large-scale energy storage deployment and because credible secondary data sources were available for both systems. The assessment is qualitative and comparative. It was intended to support broader LCSA conclusions rather than generate a formal quantitative social performance score. The study is limited to a cradle-to-grave overview using secondary literature and does not include stakeholder interviews, field surveys, or primary data collection.

Stakeholder Category	Impact Subcategory	Relevance to BESS	Relevance to PHES
Workers	Health and Safety	Occupational risks associated with lithium, cobalt, nickel, and graphite mining, battery manufacturing, and recycling	Construction hazards, tunneling risks, dam construction injuries, and long-term maintenance safety
Local Community	Access to Material Resources / Community Well-Being	Community disruption and environmental justice concerns near mining regions	Land displacement, ecosystem disruption, reservoir flooding, and local infrastructure impacts

Table 3: Selected Stakeholder Categories and Impact Subcategories

The Workers stakeholder category was selected because both technologies rely on labor-intensive extraction, manufacturing, and construction processes that expose workers to occupational safety risks. The Local Community stakeholder category was selected because both systems can significantly affect nearby communities through land use change, resource extraction, environmental disruption, and infrastructure development.

Phase 3, Impact Assessment

Worker Health and Safety

Stakeholder Category	Impact Subcategory	System	Status / Assessment	Supporting Sources
Workers	Health and Safety	BESS	Lithium-ion battery systems involve worker exposure risks associated with lithium, cobalt, and nickel mining, including hazardous dust exposure, chemical handling, and unsafe mining conditions in some global supply chains. Battery manufacturing also involves chemical exposure and high-temperature industrial processing. Injury and fatality rates in cobalt mining regions, particularly artisanal mining operations in the Democratic Republic of Congo, remain a major concern.	Amnesty International (2016); UNEP/SETAC (2013); IEA (2023)
Workers	Health and Safety	PHES	Pumped hydro systems involve major construction activities including tunneling, excavation, dam construction, and heavy equipment operation. Occupational risks are concentrated primarily during construction phases rather than distributed continuously throughout operation. Long-term operation generally involves fewer	ILO (2021); UNEP/SETAC (2013); Yang & Jackson (2011)

			hazardous material exposures compared to battery systems.	
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Table 4: Workers Stakeholder Category: Health and Safety

The worker health and safety impacts differ substantially between the two technologies. For BESS systems, worker risks are distributed across the entire supply chain, especially during mining and mineral processing. Lithium, cobalt, and nickel extraction often occur in regions with inconsistent labor protections and environmental regulations, increasing concerns regarding worker injury, hazardous exposure, and unsafe labor practices (Amnesty International, 2016). In addition, battery manufacturing introduces chemical and thermal hazards associated with large-scale industrial processing.

For PHES systems, worker risks are concentrated primarily during infrastructure construction. Excavation, tunneling, dam construction, and heavy machinery operation create serious short-term occupational hazards, although these risks become much smaller once facilities enter long-term operation.

The social tradeoff between the systems is largely tied to distribution and duration. BESS systems create ongoing supply-chain-related labor burdens across global mining and manufacturing networks, while PHES systems concentrate hazards near the construction site during shorter periods.

Local Community

Stakeholder Category	Impact Subcategory	System	Status / Assessment	Supporting Sources
Local Community	Access to Material Resources / Community Well-Being	BESS	Lithium, cobalt, and nickel extraction can create significant local environmental justice concerns, including water depletion, contamination risks, land degradation, and disruption of nearby communities. Mining-intensive regions may experience unequal economic benefits and environmental burdens.	Amnesty International (2016); UNEP (2021); IEA (2023)

Local Community	Access to Material Resources / Community Well-Being	PHES	Pumped hydro projects may require large reservoirs and significant land transformation, potentially causing displacement of local communities, habitat disruption, and changes to water availability. However, some projects may also create long-term infrastructure investment and regional employment opportunities.	UNEP/SET AC (2013); Yang & Jackson (2011); IRENA (2022)
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Table 5: Local Community Stakeholder Category: Community Well-Being and Access to Resources

For BESS systems, community impacts are closely tied to mining operations and material extraction. Lithium extraction in water-scarce regions and cobalt mining in developing economies can contribute to water depletion, land degradation, contamination risk, and unequal distribution of environmental burdens (UNEP, 2021).

For PHES systems, impacts are more localized around the project site. Reservoir development can require land transformation, ecosystem disruption, and possible community displacement. However, PHES projects can also create long-term infrastructure investment and regional employment opportunities.

Compared to BESS systems, PHES impacts are generally easier to identify geographically because they are concentrated around a single project site. BESS supply-chain impacts are distributed globally across multiple mining and manufacturing regions.

Phase 4, Interpretation

Conclusions

The Social Life Cycle Assessment showed that BESS and PHES systems create different types of social burdens across their life cycles.

For BESS systems, the largest concerns are tied to mining and supply chains. Lithium, cobalt, and nickel extraction can expose workers to hazardous conditions and can create environmental justice concerns in nearby communities, particularly in regions with weak labor protections or inconsistent environmental oversight (Amnesty International, 2016). These impacts are geographically distributed across global supply chains and continue throughout extraction, manufacturing, and recycling.

PHES systems instead concentrate social impacts near the project site. Construction activities introduce worker safety risks related to excavation, tunneling, dam construction, and heavy equipment operation. Reservoir creation can also disrupt nearby communities through land transformation and possible displacement.

The analysis conveys that neither technology can be considered universally superior from a social sustainability perspective. Instead, the socially preferable option depends heavily on project location, governance quality, labor protections, environmental safeguards, and community engagement practices. Decision-makers selecting between technologies should therefore consider not only technical and economic performance, but also the geographic distribution and duration of social impacts across the system life cycle.

In practice, policymakers may prefer BESS systems when flexible deployment and rapid renewable integration are priorities, but stronger supply-chain oversight and responsible sourcing standards would be necessary to reduce mining-related social harms. PHES systems may be more socially favorable in regions where reservoir impacts can be carefully managed and where long-term infrastructure investment provides broader community benefits.

Completeness of Study

This study relied entirely on secondary literature and publicly available reports rather than primary stakeholder interviews or field data collection. Social indicators also vary substantially across geographic regions and project implementations, which makes generalized comparison difficult.

The analysis focused specifically on two impact subcategories and therefore does not capture all possible stakeholder concerns such as cultural heritage, indigenous rights, corruption, child labor, or access to energy.

Limitations

Several important limitations affect this S-LCA. First, there is currently no universally standardized quantitative methodology for S-LCA equivalent to environmental midpoint calculations used in traditional E-LCA frameworks (UNEP/SETAC, 2013). Many social indicators remain qualitative, context-dependent, and difficult to compare directly across technologies. Second, the study is not peer-reviewed and relies on aggregated secondary data from international reports and published literature. Data availability also varies substantially between technologies and geographic regions. For example, worker safety data from artisanal

mining operations may be incomplete or inconsistently reported, while community impacts associated with future PHES projects depend heavily on site-specific geography and governance.

Finally, the analysis also evaluated generalized system types rather than project-specific case studies. Actual social outcomes would vary substantially depending on labor standards, environmental regulation, community engagement practices, and geographic context.

Response to Review, SLCA

Comments	Responses
1) Peer comment: The reviewer noted that the conclusions effectively synthesized the tradeoffs between ongoing battery impacts and concentrated hydro impacts. They suggested improving the goal statement and limitations discussion by explicitly defining what the study would not cover, including the absence of primary data collection and the cradle-to-grave overview scope.	This was a valuable comment because it identified an important clarification regarding study boundaries and methodological scope. In response, I revised the introductory inventory section and limitations discussion to explicitly state that the S-LCA is a cradle-to-grave overview based entirely on secondary literature sources and does not include primary stakeholder interviews, field surveys, or formal quantitative S-LCA modeling. I also clarified that the study evaluates generalized technology categories rather than project-specific case studies. These additions improve transparency regarding the scope and limitations of the analysis.
2) Peer comment: The reviewer appreciated the discussion of stakeholder tradeoffs and use of credible references such as Amnesty International, UNEP, and ISO-related guidance. They suggested more directly linking limitations to specific stakeholders and life cycle stages, while also discussing situations in which policymakers may still choose each technology and what safeguards would be necessary.	This comment helped strengthen the interpretation section by connecting social tradeoffs more directly to real-world decision-making. In response, I expanded the conclusions section to explain situations where policymakers or planners may still prefer either BESS or PHES systems depending on project goals and geographic conditions. I also added discussion about safeguards such as responsible sourcing standards, labor protections, environmental oversight, and community engagement practices. In addition, I revised the limitations section to more explicitly identify how data gaps affect different stakeholders and life cycle stages differently across the two technologies.

3) TA comment: The TA noted that the analysis was very thorough and suggested strengthening the comparison further by incorporating more direct quantitative comparisons within the discussion sections. Another TA comment suggested adding a fifth column identifying the broader impact category in the assessment tables.	In response to these comments, I revised the impact assessment tables to more clearly structure stakeholder categories and impact subcategories while strengthening the comparative discussion language throughout the narrative. I added more direct comparisons between geographically distributed mining-related impacts for BESS systems and localized construction-related impacts for PHES systems. I also incorporated clearer references to worker injury risks, environmental justice concerns, and infrastructure-related disruptions within the interpretation paragraphs so that the implications are easier to compare directly across systems. Due to inconsistent international reporting standards and limited directly comparable datasets, fully quantitative comparison metrics were beyond the scope of this project, but future work could incorporate standardized injury-rate or community-impact indicators where available.
4) Self comment: While revising the S-LCA section, I realized that the earlier draft focused heavily on describing stakeholder impacts individually but did not fully explain how governance quality and regional context influence whether these impacts become more or less severe in practice.	To improve this, I expanded the conclusions and limitations sections to discuss how labor protections, environmental regulations, mining oversight, and community engagement practices can significantly alter social outcomes for both technologies. I also added discussion explaining that the social sustainability of each system depends heavily on implementation context rather than the technology alone. If additional time and resources were available, future work could incorporate region-specific case studies, stakeholder interviews, and more formal social indicator scoring methods to improve comparability and depth.

Conclusions & Recommendations

Overall, this LCSA does not support a simple conclusion that either lithium-ion BESS or PHES is always the more sustainable energy storage option. Each system performs differently depending on which sustainability category is prioritized.

From the Environmental LCA, BESS systems are dominated by upstream burdens from mineral extraction and battery manufacturing, while PHES systems are dominated by construction-stage impacts tied to concrete production, excavation, reservoir flooding, and land transformation (IEA, 2023; Yang & Jackson, 2011).

From the LCC analysis, PHES generally performs better for long-duration storage because long operational lifetimes reduce replacement-related costs over time (IRENA, 2022). BESS systems are easier to deploy and less geographically constrained, but battery degradation and replacement significantly increase long-term cost.

From the S-LCA, BESS systems raise larger concerns related to mining, worker protections, and environmental justice across global supply chains. PHES systems instead concentrate social impacts near the project site through construction hazards, land use change, and possible community disruption.

The final recommendation is that PHES is the stronger sustainability option for large-scale, long-duration storage where suitable geography exists and ecosystem impacts can be responsibly managed. BESS remains valuable for modular, flexible, and shorter-duration storage applications where rapid deployment and siting flexibility are priorities.

The largest limitation across the study is data consistency. Environmental, cost, and social data were drawn from different sources, regions, assumptions, and project scales. Because of this, the results should be interpreted as a comparative screening-level assessment rather than a definitive ranking. Future work should incorporate project-specific inventory data, regional electricity mixes, discounted cash flow modeling, and more detailed social data for mining communities and PHES host regions.

References

Amnesty International. (2016). *This is what we die for: Human rights abuses in the Democratic Republic of the Congo power the global trade in cobalt*. Amnesty International.

<https://www.amnesty.org/en/>

Franze, J., & Ciroth, A. (2011). A comparison of cut roses from Ecuador and the Netherlands. In United Nations Environment Programme (UNEP), *Towards a life cycle sustainability assessment: Making informed choices on products* (pp. 373–378). UNEP.

International Energy Agency. (2023). *Batteries and secure energy transitions*. IEA.

<https://www.iea.org/reports/batteries-and-secure-energy-transitions>

International Labour Organization. (2021). *Safety and health in construction*. ILO.

<https://www.ilo.org/resource/other/safety-and-health-construction-revised-edition>

International Renewable Energy Agency. (2022). *Electricity storage and renewables: Costs and markets to 2030*. IRENA.

<https://www.irena.org/publications/2017/Oct/Electricity-storage-and-renewables-costs-and-markets>

Lazard. (2024). *Levelized cost of storage analysis, version 10.0*. Lazard.

<https://www.lazard.com/media/5tlbhyla/lazards-lcoeplus-june-2025-vf.pdf>

National Renewable Energy Laboratory. (2022). *Cost projections for utility-scale battery storage*.

U.S. Department of Energy. <https://docs.nrel.gov/docs/fy25osti/93281.pdf>

U.S. Department of Energy. (2023). *Grid energy storage technology cost and performance assessment*. U.S. Department of Energy, Office of Electricity.

<https://www.energy.gov/oe/oe-reports-and-other-documents>

U.S. Department of Energy. (2023). *Long duration storage shot*. U.S. Department of Energy.

<https://www.energy.gov/cmei/oced/long-duration-energy-storage>

United Nations Environment Programme. (2021). *Mineral resource governance in the 21st century: Gearing extractive industries towards sustainable development*. UNEP.

<https://www.unep.org/resources/report/mineral-resource-governance-21st-century>

United Nations Environment Programme. (2011). *Towards a life cycle sustainability assessment: Making informed choices on products*. UNEP.

<https://www.unep.org/resources/report/towards-life-cycle-sustainability-assessment-making-informed-choices-products>

United Nations Environment Programme & Society of Environmental Toxicology and Chemistry. (2013). *The methodological sheets for sub-categories in social life cycle assessment (S-LCA)*. UNEP/SETAC Life Cycle Initiative.

Yang, C. J., & Jackson, R. B. (2011). Opportunities and barriers to pumped-hydro energy storage in the United States. *Renewable and Sustainable Energy Reviews*, 15(1), 839–844.
<https://ideas.repec.org/a/eee/rensus/v15y2011i1p839-844.html>

Zakeri, B., & Syri, S. (2015). Electrical energy storage systems: A comparative life cycle cost analysis. *Renewable and Sustainable Energy Reviews*, 42, 569–596.
<https://www.sciencedirect.com/science/article/abs/pii/S1364032114008284>